WORKED SOLUTION

STEP III Specimen paper, Question 16

Probability and Statistics • Negative Binomial Distribution

STEP III Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q16

The key idea

Waiting for r sons is the same as waiting for one son, then starting again and waiting for the next, r times over. So C splits into a sum of r identical waiting times, and linearity of expectation does the rest. For part (ii), the trick is to imagine the births carrying on past the point where the president stops: whether he has stopped by a given time is then a question about a *fixed* number of children, and a symmetry between boys and girls settles it.

Solution

Throughout, each birth is a boy with probability $\frac{1}{2}$, independently of all other births.

Part (i)

For $i = 1, 2, \ldots, r$, let X_i be the number of children born after the $(i-1)$th son, up to and including the ith son. (For $i = 1$, count from the start.) Then

$$C = X_1 + X_2 + \cdots + X_r.$$

Each X_i counts births up to the first boy in a fresh run of independent fair births, so every X_i has the same distribution as X, where

$$\mathrm{P}(X = k) = \left(\tfrac{1}{2}\right)^{k-1} \cdot \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^{k}, \qquad k = 1, 2, 3, \ldots,$$

since $X = k$ means $k - 1$ girls followed by a boy.

To find $\mathrm{E}(X)$, differentiate the geometric series $\sum_{k\geq 0} x^k = \dfrac{1}{1-x}$ for $|x| < 1$ to get $\sum_{k\geq 1} kx^{k-1} = \dfrac{1}{(1-x)^2}$. Multiplying by x and putting $x = \frac{1}{2}$,

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} k\left(\tfrac{1}{2}\right)^{k} = \frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)^2} = 2.$$

This makes sense: if half of all babies are boys, you expect to wait two births for one.

By linearity of expectation (which holds whether or not the X_i are independent, though here they are),

$$\mathrm{E}(C) = \mathrm{E}(X_1) + \cdots + \mathrm{E}(X_r) = 2r.$$

Part (ii)

Imagine that the sexes of an endless sequence of babies are decided in advance, independently and fairly, and that the president has babies from this sequence until he reaches his rth son. This changes nothing about C, but it lets us talk about "the first $2r-1$ babies" even when he stops sooner.

Ask yourself How can the event $C < 2r$ be described using only the first $2r-1$ babies of the sequence?

Since C is a whole number, $C < 2r$ means $C \leq 2r-1$, that is, the rth son arrives at or before position $2r-1$. That happens exactly when there are at least r boys among the first $2r-1$ babies. So

$$\mathrm{P}(C < 2r) = \mathrm{P}(\text{at least } r \text{ boys among the first } 2r-1 \text{ babies}).$$

Now let B and G be the numbers of boys and girls among these $2r-1$ babies, so $B+G = 2r-1$. Because $2r-1$ is odd, B and G cannot be equal, and exactly one of them is at least r:

- if $B \geq r$ then $G = 2r-1-B \leq r-1$;
- if $B \leq r-1$ then $G \geq r$.

So the events $\{B \geq r\}$ and $\{G \geq r\}$ are mutually exclusive and between them cover every possibility, which gives

$$\mathrm{P}(B \geq r) + \mathrm{P}(G \geq r) = 1.$$

But boys and girls are equally likely at every birth, so swapping the labels "boy" and "girl" doesn't change any probability, and $\mathrm{P}(B \geq r) = \mathrm{P}(G \geq r)$. Hence each is $\frac{1}{2}$, and

$$\mathrm{P}(C < 2r) = \tfrac{1}{2},$$

as required.

So the expected family size is $2r$, yet the family is smaller than $2r$ exactly half the time. The family sizes are skewed: values just below $2r$ are common, and the occasional very long run of girls pulls the mean up.

Checks

- With $r = 1$: $\mathrm{E}(C) = 2$, and $C < 2$ means the first baby is a boy, which has probability $\frac{1}{2}$.

- With $r = 2$: $C < 4$ means $C = 2$ or 3, with probabilities $\frac{1}{4}$ and $2 \cdot \frac{1}{8} = \frac{1}{4}$, total $\frac{1}{2}$.
- Summing $c\binom{c-1}{r-1}2^{-c}$ exactly for $r = 1, \ldots, 8$ gives $\mathrm{E}(C) = 2r$ and $\mathrm{P}(C < 2r) = \frac{1}{2}$ every time; a simulation with $r = 3$ and 200 000 families gave a mean of 6.00 and $\mathrm{P}(C < 6) \approx 0.499$.

Common slips

- Arguing "half the children are boys, so $C = 2r$". That is true on average, but C itself is random; part (i) needs the decomposition into waiting times, not just a guess.
- Considering only the president's actual children in part (ii). When $C < 2r - 1$ there *are* no $2r - 1$ children; extending the sequence of births past the stopping point is what makes the counting argument valid.
- Forgetting that $2r - 1$ is odd. That is exactly what rules out a tie between boys and girls and makes the two events complementary.

Beyond STEP

The distribution of C is the *negative binomial* distribution: $\mathrm{P}(C = c) = \binom{c-1}{r-1}2^{-c}$ for $c \geq r$. Writing a random variable as a sum of simpler pieces and using linearity of expectation is one of the most powerful tools in first-year probability, and the trick in part (ii) of linking a "waiting time" to a "count in a fixed number of trials" is the same duality that connects the negative binomial and binomial distributions (and, in continuous time, the gamma and Poisson distributions). Blitzstein and Hwang, *Introduction to Probability*, treats both ideas with many examples.

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